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CAPSTONE PROJECT

Closed-Loop Medication & Pharmacy Optimization

Apex Health Network (AHN)

POD Name: POD ORION 1

Programme: Health Tech Business Analysis

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Business case reference: AHN-CLM-BC-001

Version: 1.0 — for academic review and executive consideration

Date: 9 October 2026

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1. The Current Challenge

A fragmented medication workflow across people, paper and disconnected systems.

12 inpatient hospitals

A network-wide medication process to improve.

45 outpatient clinics

Broader network; exact rollout scope to confirm.

Manual hand-offs

Paper charts, verbal checks and legacy inventory tools.

Potential business and clinical impact

- Medication errors and delays in verification or administration
- Stockouts, expiry waste and avoidable nursing workload
- Incomplete billing, weak audit trails and limited management insight

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2. Why Change Is Needed

Root causes span technology, process, people, governance and data.

Technology

Disconnected systems and limited interoperability.

Process

Manual hand-offs and duplicated checks create delays and rework.

People & adoption

Usability, workload and training gaps can encourage workarounds.

Governance & data

Inconsistent standards and fragmented data weaken oversight.

The case for change: connect the full medication journey—not just one isolated system.

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3. Proposed Digital Solution

A single connected workflow—from prescribing through reconciliation.

1 PRESCRIBE
Electronic order

→

2 VERIFY
Clinical + pharmacist checks

→

3 DISPENSE
Prepare medicine + confirm stock

→

4 ADMINISTER
Barcode scan + eMAR record

→

5 RECONCILE
Update stock, billing + audit trail

CONNECTED CAPABILITIES

Inventory intelligence
Stock availability, expiry and replenishment alerts

Controlled medicines
Traceable transactions and audit trail

Operational analytics
Safety, waste and billing KPIs

Enter once → Verify safely → Record every step → Reconcile automatically

4. Future-State Digital Workflow

A closed-loop process connects clinical decisions, pharmacy operations and the patient record.

01
ORDER
Prescriber enters order in EHR/HIS

02
CLINICAL CHECK
Pharmacist verifies dose, allergies and interactions

03
STOCK CHECK
System checks availability, expiry and replenishment

04
DISPENSE
Pharmacy prepares and records the medication

05
BEDSIDE SCAN
Staff scan patient + medicine; right-patient checks

06
ADMINISTER
Administration is recorded in eMAR

07
CLOSE THE LOOP
Inventory and billing update; exceptions feed dashboards

Exceptions → review → documented resolution

Safety gate: unresolved clinical alerts require authorised review before administration.

5. AS-IS to TO-BE: The Future State

Retain the problems already identified—and show the specific change the digital solution enables.

AS-IS | CURRENT PROBLEMS		TO-BE | FUTURE STATE
Paper charts, verbal checks and manual hand-offs	→	A shared electronic medication record with time-stamped actions
Disconnected legacy tools and duplicated data entry	→	EHR/HIS-integrated workflow with information recorded once
Medication verification and administration delays	→	Structured pharmacist review and barcode-supported bedside checks
Limited visibility of stock, expiry and replenishment	→	Near-real-time inventory visibility with stock and expiry alerts
Weak audit trails and incomplete billing reconciliation	→	Traceable dispense/administer events linked to inventory and billing
Fragmented reporting and limited management insight	→	Dashboards for safety, stockouts, waste, adoption and billing KPIs

Expected direction: safer administration • less avoidable waste • stronger traceability • clearer oversight

Targets remain proposed until AHN baselines are measured and validated.

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4. What Success Should Look Like

Targets from the write-up are proposed outcomes; baselines must be measured first.

−40% **Medication-administration errors** End of Year 3	**−50%** **Critical stockout events** End of Year 2	**−35%** **Expiry / waste cost** End of Year 3
≥95% **BCMA/eMAR adoption** End of Year 2	**≥98%** **Inventory accuracy** End of Year 2	**≤15%** **Shift time on manual checking** End of Year 2

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5. Stakeholders & Governance

Clinical safety and frontline adoption are as important as technology delivery.

Manage closely

Sponsor, CMO, CNO, Chief Pharmacist, IT and patient-safety leads.

Co-design & test

Doctors, nurses, pharmacists, inventory, finance and patient representatives.

Governance controls

- Maintain stakeholder register, RACI, decision and risk/issue logs
- Formal clinical-safety, privacy, cybersecurity and architecture approvals
- Train staff; test usability and downtime procedures; track go-live readiness

6. Delivery Roadmap

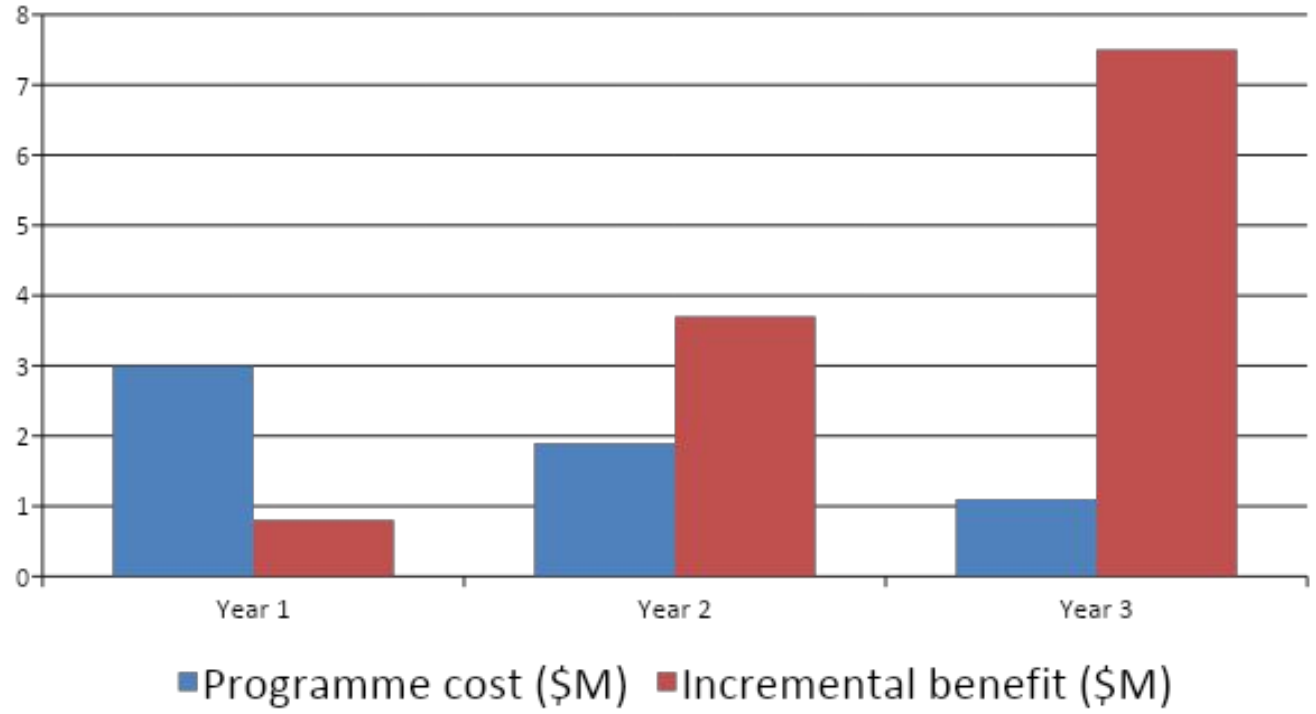
Hybrid Agile: iterative co-design within formal clinical and technical stage gates.

PHASE 1	Months 0–12	**Discover, design & pilot**	Integrations, eMAR/BCMA, baseline measures, UAT and rollback proof.
PHASE 2	Months 13–24	**Scale inpatient rollout**	Inventory, billing, controlled medicines and dashboards.
PHASE 3	Months 25–36	**Optimise & extend**	Advanced analytics and selected outpatient workflows.

Scale only after safety sign-off, User Acceptance Test (UAT), training, downtime rehearsal and rollback readiness.

7. Illustrative Financial Case

Planning assumptions only—validate with AHN Finance, baselines and vendor quotes.



Total programme cost

$6.0M

Total benefits

$12.0M over 3 years

Net benefit

$6.0M illustrative

- **ROI: 100%**
- **Benefit Cost Ratio: 2.0×**
- **Payback: ≈2.1 years**
- **Net Present Value: ≈$4.59M**

8. Key Risks & Safeguards

Build safety, resilience, privacy and adoption into delivery from day one.

Clinical workflow risk

Clinical validation, pharmacist checks, BCMA rules and a signed safety case.

Integration & downtime

Early interface testing, high availability and tested rollback procedures.

Privacy & cybersecurity

Least-privilege access, authentication, encryption, audit logs and testing.

Low adoption / weak benefits

Frontline co-design, super-users, competency checks and named KPI owners.

No scale-up without approved gates, UAT and evidence of readiness.

9. Recommendation & Decision Requested

Approve Phase 1 discovery and pilot planning—not unconditional full rollout.

Approve Phase 1

Authorise discovery, baseline measurement, requirements and pilot planning.

Confirm accountable owners

Sponsor, clinical safety, pharmacy, nursing, IT, privacy and Finance.

Before the pilot

- Validate costs, benefits and operational baselines
- Complete architecture, privacy/security and clinical-safety reviews
- Pass UAT; train staff; prove downtime and rollback procedures

Decision: proceed to a controlled pilot, then scale only when evidence supports it.